

Final System Vision

AI Parking Intelligence Platform

Input:

Violation Records

+

GPS Location

+

Vehicle Type

+

Violation Type

+

Junction Information

+

Timestamp

Output:

Hotspot Detection

Risk Scoring

Future Risk Prediction

Patrol Allocation

Impact Simulation

Data Filtering (Before Module 1)

This needs to be explicitly defined.

Your EDA showed:

approved = 115,400

NaN = 125,254

rejected = 49,754

duplicate = 320

Recommended Dataset

Use:

```
valid_df = df[(
    df["validation_status"].isna()
)
|
(
    df["validation_status"] == "approved"
)
]
```

Exclude:

rejected

duplicate

Reason:

- rejected = known bad
- duplicate = known bad
- NaN ≠ necessarily invalid

This gives maximum coverage while remaining defensible.

Module 1 — Spatial Hotspot Detection

Goal

Identify illegal parking concentration zones.

IMPORTANT CHANGE

Do NOT run:

```
DBSCAN(metric="euclidean")
```

Use:

```
DBSCAN(  
    metric="haversine",  
    algorithm="ball_tree"  
)
```

because latitude/longitude are geographic coordinates.

Inputs

latitude

longitude

Outputs

cluster_id

cluster_density

hotspot_rank

Deliverables

- Heatmap
- Cluster map
- Top 20 hotspot zones

Module 2 — Temporal Intelligence

Goal

Understand enforcement and violation patterns.

Features

hour

day_of_week

month

is_weekend

Key Insight

Do NOT claim:

Peak traffic hour

Instead say:

Peak detection hour

because your EDA strongly suggests timestamps reflect enforcement activity rather than actual parking start times.

Outputs

Detection-hour distribution

Weekend trends

Zone-wise active windows

Module 3 — Congestion Risk Engine

This is the core innovation.

A. Violation Severity

Example:

severity_weight = {

 "NO PARKING": 1,

 "WRONG PARKING": 2,

```
"PARKING ON FOOTPATH": 3,  
"DOUBLE PARKING": 4,  
"PARKING IN A MAIN ROAD": 5,  
"PARKING NEAR ROAD CROSSING": 6,  
"PARKING NEAR TRAFFIC LIGHT OR ZEBRA CROSS": 7  
}
```

B. Vehicle Size Weight

This was an excellent addition from the review.

Example:

```
vehicle_weight = {  
  "SCOOTER": 1.0,  
  "MOTORCYCLE": 1.0,  
  "CAR": 1.5,  
  "AUTO": 1.8,  
  "LGV": 2.5,  
  "TEMPO": 2.5,  
  "BUS": 3.5,  
  "LORRY": 4.0,  
  "HGV": 4.0,  
  "TANKER": 4.0  
}
```

This directly relates to road space occupied.

C. Junction Importance

Do NOT use:

junction_flag = 0/1

Use:

junction_weight

derived from:

historical junction violation counts

Example:

Safina Plaza

KR Market

Elite Junction

receive higher weights than low-volume junctions.

D. Multi-Violation Bonus

You discovered:

single

double

triple

violations.

Use:

multi_violation_weight

E. Persistence Score

For each hotspot:

months_active

This is one of the strongest operational features.

Final Priority Score

priority_score =

0.35 * density_score

+ 0.20 * violation_severity

+ 0.15 * vehicle_weight

+ 0.15 * junction_weight

+ 0.10 * persistence_score

+ 0.05 * multi_violation_score

Congestion Disclaimer

Use this exact narrative:

Direct traffic-flow measurements are unavailable in the dataset. Therefore, congestion impact is estimated using a surrogate model combining violation density, violation severity, vehicle size, hotspot persistence, and junction importance.

This protects you during judging.

Module 4 — Risk Prediction

Model

XGBoost Classifier

Not regression.

Predict

Low Risk

Medium Risk

High Risk

Critical

Features

cluster_id

hour

day_of_week

month

vehicle_type

violation_type

junction_weight

priority_score

Output

Risk level for hotspot

Probability score

Module 5 — Patrol Allocation Engine

Input:

Current Risk

Predicted Risk

Output:

Top patrol zones

Suggested patrol windows

Priority ranking

Example:

Zone 12

Critical Risk

Suggested Monitoring:

4 AM – 6 AM

Module 6 — Impact Simulator

Keep it simple.

Single slider:

Violation Reduction %

Example:

Current Priority Score = 91

Reduce violations by 30%

New Priority Score = 63

Do not claim actual traffic reduction.

Only claim:

Predicted hotspot risk reduction

Dashboard

Tab 1

City Hotspot Map

Tab 2

Top Risk Zones

Tab 3

Temporal Patterns

Tab 4

Risk Forecasts

Tab 5

Patrol Recommendations

Tab 6

Impact Simulator

Immediate Next Step

Do **not** start XGBoost yet.

The next task is:

1. Apply validation-status filtering
2. Convert coordinates to radians
3. Run Haversine DBSCAN
4. Generate cluster IDs

5. Compute hotspot statistics

Everything else depends on `cluster_id`.

Without clusters:

- no persistence score
- no hotspot ranking
- no risk prediction
- no patrol optimizer

So the project is now at:

EDA 

Data Filtering 

DBSCAN Hotspot Generation 

Feature Engineering 

XGBoost 

Dashboard 